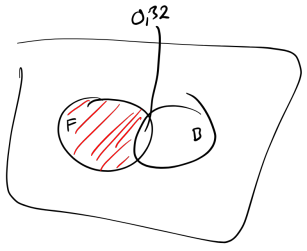
$P(B \cap F) = 0{,}32$

$P(B' \cap F) = 0{,}24$

$P(F' \cap B) = 0{,}1$



$P(F) = 0{,}32 + 0{,}24$

$P(B) = 0{,}24 + 0{,}1$

$P(B|F) = \frac{0{,}32}{0{,}56}$

Opg 1

Opg 2

a)

$P(A) = 0{,}97$

$P(B) = 0{,}93$

$P(A' \cap B') = P((A \cap B)')$

Uavhengig $P(A' \cap B') = P(A') \cdot P(B')$

$= 0{,}03 \cdot 0{,}07$

$= 0{,}0021$

b) $P(B|A) = \frac{P(B \cap A)}{P(A)}$

Uavhengighet:

$P(B|A) = P(B)$

Opg 3

C = smitte

T = pos. test

$P(T|C) = 0{,}77$

$P(T'|C') = 0{,}99$

$P(C) = 0{,}03$

a) $P(T|C') = 1 - P(T'|C')$

$= 1 - 0{,}99$

$= 0{,}010$

$P(T|C') = \frac{P(C'|T)\,P(T)}{P(C')} = \frac{(1 - P(C|T))\,P(T)}{1 - P(C)}$

$P(T)$: $P(C|T) = \frac{P(T|C)\,P(C)}{P(T)} \Rightarrow P(T) = \frac{P(T|C)\,P(C)}{P(C|T)}$

$$\frac{(1 - P(C|T))\,\frac{P(T|C)\,P(C)}{P(C|T)}}{1 - P(C)} = \frac{\frac{P(T|C)\,P(C)}{P(C|T)} - P(T|C)\,P(C)}{1 - P(C)}$$

$$\frac{P(T) - 0{,}77 \cdot 0{,}03}{0{,}97}$$

$P(T) = P(T|C')P(C') + P(T|C)P(C)$

$$P(c|T) = 1 - P(c'|T)$$

$$P(c'|T) = \frac{P(T|c')\,P(c')}{P(T)} = \frac{0,01 \cdot 0,97}{0,0328}$$

$$P(c|T') = (\text{se notisbok}) = \frac{(1 - P(T|c))P(c)}{1 - P(T)}$$

$$= \frac{(1-0,77)0,03}{1-0,0328} = \frac{0,0069}{0,9672} = 0,007$$

Oppg 4

X = # ganger terning 1

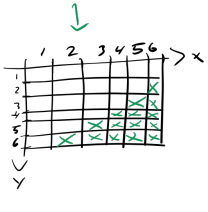
Y = # ganger terning 2

Z = X + Y

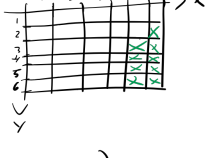
X ∩ Y = overlappende

$$P(X \geq 5) = \frac{g}{m} = \frac{2}{6} = 0,\overline{33}$$

$$P(Z \geq 8) = \frac{g}{m} = \frac{\quad}{36}$$



$P(Z \geq 8 \cap X \geq 5)$



$$P(Z \geq 8 \mid X \geq 5) = \frac{P(Z \geq 8 \cap X \geq 5)}{P(X \geq 5)} = \frac{1/4}{1/3} = \frac{1}{4}\frac{3}{1} = \frac{3}{4}$$

$P(Z \geq 8 \cap X \geq 5) \overset{?}{=} P(Z \geq 8) \cdot P(X \geq 5)$

$\frac{9}{36} \overset{?}{=} \frac{15}{36} \times \frac{1}{3}$

$9 \neq 5$

ikke uavhengig

Oppg 5

$f(x) = \begin{cases} 4x^3, & 0 \leq x \leq 1 \\ 0, & \text{ellers} \end{cases}$, $\bar{X} =$ $\frac{\text{oppnådd batterikap. etter 100 anrm}}{\text{alle disker}}$

a) $F(X \leq x) = \int f(x)\,dx = x^4$ for $0 \leq x \leq 1$

b) $k = ?$ $P(X > k) = \frac{1}{8} = P(1 \geq X \geq k)$

$$\int_k^1 4x^3\,dx = \frac{1}{8}$$

$$1^4 - k^4 = \frac{1}{8}$$

$$\sqrt[4]{1 - 0{,}125} = k$$

$$k = 0{,}967$$

c) $P(\frac{1}{5} \leq X \leq 2) = P(\frac{1}{5} \leq X \leq 1) = \int_{1/5}^{1} f(x)\,dx = 1 - \frac{1}{5^4} = 0{,}9984$

Oppg 6

$\underline{X}$ og $\underline{Y}$ = kontinuerlig stokastisk variabel

$$f_{\underline{X},\underline{Y}}(x,y) = \begin{cases} \frac{2(4y+x)}{5}, & 0 \le x, y \le 1 \\ 0, & \text{ellers} \end{cases}$$

= simultantetthet

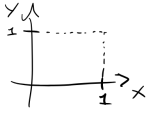
$f_{\underline{X}}(x) = ?$ = Marginaltetthet

$$f_{\underline{X}}(x) = \sum_y f_{xy}(x,y)$$

$$f_{\underline{X}}(x) = \int_0^1 f_{xy}(x,y)\,dy = \frac{2}{5}\int_0^1 4y+x\,dy = \frac{2}{5}\left[2y^2+xy\right]_0^1$$

$$= \frac{2}{5}(2+x)$$

b) $\underline{Y} \mid \underline{X} = x$: $f_{\underline{Y}|x=x}(y|x) = \dfrac{f_{xy}(x,y)}{f_X(x)}$



$$= \frac{\frac{2}{5}(4y+x)}{\frac{2}{5}(2+x)}$$

$$= \frac{4y+x}{2+x}$$